

Guide 30 - Electrician and Electrical Technician

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Purpose: Practical career-entry and advancement guide for electricians and closely related electrical-service, maintenance, construction, and technician roles.

This guide is educational. It does not guarantee employment, wages, admission, funding, reimbursement, apprenticeship placement, certification, licensing, promotion, or legal authority to perform electrical work. Electrical work is safety-critical and may be regulated by state, provincial, territorial, municipal, national, employer, site, utility, or other rules. Verify current requirements before paying for training or performing regulated or hazardous work.

1. What this work is and who it may fit

Electricians install, maintain, troubleshoot, and repair electrical power, lighting, communications, control, and related systems. Work may occur in homes, commercial buildings, industrial facilities, construction projects, public infrastructure, institutional facilities, renewable-energy projects, manufacturing plants, utilities, and maintenance departments.

The occupation can fit people who enjoy practical problem-solving, precise measurement, diagrams, tools, mathematics, systems thinking, troubleshooting, and visible results. It also requires disciplined safety behavior. Electrical energy can cause shock, burns, arc-flash injury, fire, explosion, falls, equipment damage, or death. A person should never treat confidence, a job title, an online tutorial, or an AI answer as a substitute for training, authorization, deenergization, qualified supervision, required PPE, lockout/tagout, permits, or applicable electrical codes.

Modern electrician work increasingly overlaps with electronic controls, programmable devices, building automation, solar photovoltaic systems, energy storage, electric-vehicle charging, networked equipment, sensors, power-quality monitoring, and digital maintenance records. These developments create opportunities, but they do not erase the legal or safety boundaries of electrical work.

2. What electricians and electrical technicians actually do

Responsibilities depend on employer, trade status, work setting, jurisdiction, voltage level, licensing, and specialization. Typical electrician work may include:

- reading electrical drawings, specifications, one-line diagrams, wiring diagrams, panel schedules, equipment manuals, and work orders;
- installing raceways, boxes, cable, conductors, devices, panels, fixtures, controls, and approved electrical equipment within authorized scope;

- testing circuits and equipment with appropriate meters and test instruments;
- locating faults such as open circuits, short circuits, ground faults, damaged conductors, failed components, loose connections, control faults, or abnormal electrical conditions;
- maintaining motors, controls, lighting, distribution equipment, and building electrical systems where trained and authorized;
- replacing approved components and documenting work;
- verifying labeling, identification, grounding/bonding, protective devices, and installation conditions according to the governing requirements and employer procedures;
- coordinating with engineers, inspectors, utilities, general contractors, maintenance teams, controls specialists, HVAC technicians, solar installers, and other trades;
- supporting commissioning, preventive maintenance, shutdowns, upgrades, renovations, and emergency response within authorized scope;
- maintaining accurate records of testing, defects, repairs, permits, inspections, unresolved conditions, and required escalation.

The title **electrical technician** can mean different things. Some positions are essentially electrician roles; others focus on industrial controls, electronics, instrumentation, testing, power systems, engineering support, maintenance, or equipment repair. Always read the actual duties and occupational classification rather than assuming that every “electrical technician” job is equivalent to U.S. SOC 47-2111 Electricians.

3. Safety, energized-work, and task boundaries

Electrical work is safety-critical. The safest career guide teaches people how to recognize hazards and escalation points, not how to improvise around them.

Deenergization and safe work practices

OSHA 29 CFR 1910 Subpart S addresses electrical safety in general industry. OSHA 1910.333 states that live parts to which an employee may be exposed generally must be deenergized before work on or near them unless the employer can demonstrate a specified exception such as infeasibility or increased hazard.

Do not assume a switch, breaker position, software command, disconnect handle, indicator light, or equipment stop button proves that equipment is electrically safe. Follow the employer's approved procedure for identifying all energy sources, disconnecting or isolating energy, applying lockout/tagout when required, controlling stored energy, verifying the condition, and restricting work to qualified and authorized persons.

Qualified-person boundary

OSHA 1910.331 distinguishes qualified and unqualified persons for electrical safety-related work practices. Training, supervision, employer authorization, and the nature of the exposed hazard matter. A learner, helper, trainee, or new employee must not infer “qualified person” status merely from completing an online course or understanding theory.

Construction, industrial, utility, and special environments

Different standards may apply depending on the work. General-industry electrical rules, construction standards, utility/power-generation requirements, lockout/tagout, fall protection, confined spaces, PPE, hazardous locations, and site-specific procedures may all become relevant.

Stop and escalate when

- the circuit, voltage, energy source, grounding condition, isolation point, or equipment state is uncertain;
- deenergization or lockout/tagout cannot be completed according to the approved procedure;
- energized work is proposed without the employer's required authorization, justification, PPE, procedures, boundaries, and qualified personnel;
- required drawings, labels, test instruments, calibration status, PPE, barriers, permits, or manufacturer instructions are missing;
- the work requires a license, registration, permit, inspection, trade status, certification, or supervision that you do not have;
- the measured condition conflicts with the expected circuit behavior;
- conductors, equipment, protective devices, grounding/bonding, hazardous locations, or system ratings are outside your verified competence;
- a customer, schedule, production target, or emergency request would require bypassing a safety control or legal requirement.

No deadline justifies unsafe energized work, defeating a protective device, falsifying a test result, bypassing an inspection, or working outside verified authorization.

4. United States licensing, codes, and legal-work boundaries

There is no single U.S. national electrician license that authorizes all electrical work everywhere. Licensing and permitting may be controlled by states, territories, counties, municipalities, or other authorities. Requirements may distinguish apprentice, trainee, journeyman, residential electrician, master electrician, electrical contractor, specialty classifications, or other categories.

The U.S. Bureau of Labor Statistics advises workers to check state licensing requirements. Before relying on a training provider's claim that a course “makes you an electrician,” verify:

- which jurisdiction regulates the work;
- the exact license or registration category;
- required classroom or technical instruction;
- required supervised work hours;
- examination requirements;
- who may supervise or sign experience;
- application, background, insurance, or business requirements;
- renewal or continuing-education requirements;

- reciprocity rules, if any;
- which tasks require permits or inspections.

Electrical codes and adopted editions vary by jurisdiction. A code book or training course does not itself grant legal authority to perform work. Verify the locally adopted code, amendments, permitting authority, inspection process, and employer procedures for the job location.

5. Entry routes and transferable skills

The U.S. Bureau of Labor Statistics lists a high school diploma or equivalent as typical entry education and apprenticeship as typical on-the-job training for electricians.

Common entry routes include:

- registered trainee or apprentice pathways where available and legally recognized;
- union-management or contractor-association apprenticeship/training programs;
- employer-sponsored helper or trainee positions with structured supervision;
- public technical-school or community-college electrical programs;
- pre-apprenticeship programs connected to real employers or recognized apprenticeship systems;
- military electrical, facilities, construction, maintenance, electronics, power, or controls experience translated carefully into civilian requirements;
- adjacent maintenance, HVAC, solar, industrial, controls, or building-operations experience;
- public vocational training such as SENA electrical programs in Colombia.

Useful transferable skills include:

- mathematics and measurement;
- mechanical reasoning;
- safe hand- and power-tool use;
- reading diagrams and technical documents;
- troubleshooting discipline;
- documentation and work-order systems;
- customer and team communication;
- basic computer and digital-tool skills;
- punctuality and reliability;
- willingness to stop and ask for qualified help.

Do not claim journeyman status, master status, apprenticeship completion, CONTE registration, Red Seal endorsement, licensing, energized-work qualifications, high-voltage experience, code expertise, or years of experience you do not actually have.

6. United States labor-market evidence

Official national statistics

The U.S. Bureau of Labor Statistics reported for Electricians (SOC 47-2111):

- **2024 employment: 818,700;**
- **May 2024 median pay: \$62,350 per year / \$29.98 per hour;**
- **projected 2034 employment: 896,100;**
- **projected growth, 2024-2034: 9%;**
- **projected numeric growth: 77,400;**
- **average projected openings: about 81,000 per year** over the decade;
- **typical entry education:** high school diploma or equivalent;
- **typical on-the-job training:** apprenticeship.

These are national occupational statistics, not entry-level offers or wage guarantees. Compensation varies by jurisdiction, union coverage, construction cycle, industrial specialization, overtime, shift work, travel, licensing level, contractor status, renewable-energy work, maintenance specialization, and experience.

Clearly labeled non-government market estimate

Indeed's U.S. electrician salary page, updated July 20, 2026, reported an average base salary of approximately 31.39 *per hour*, with displayed low/high values around ***\$18.91-\$52.11 per hour** and an additional overtime indicator.

This is a **non-government market estimate** based on Indeed's methodology and job-posting/salary observations. It is not definition-equivalent to the BLS median and is not a guaranteed wage.

When comparing offers, ask how overtime, shift premiums, travel, per diem, tools, uniforms, vehicle use, training time, licensing fees, continuing education, union dues, testing costs, and standby or emergency work are handled.

7. Free and low-cost U.S. pathways

Start with paid, public, workforce-funded, union/industry, or employer-supported options before taking on private-school debt.

Public workforce and education routes

Check:

- local American Job Centers and WIOA-funded training options;
- community colleges and public technical schools;
- state apprenticeship agencies or recognized apprenticeship sponsors;
- employer helper/trainee programs;
- union-management apprenticeship committees;

- contractor-association training programs;
- high-school or adult career-technical education;
- veterans' and military-transition resources when applicable.

FAFSA, Pell Grants, and scholarships

Eligible students at participating colleges or career schools may qualify for federal student aid. Do not assume a private electrician course qualifies. Verify the institution, program, student eligibility, total cost, refund rules, completion rate, job-placement methodology, and whether the program actually satisfies licensing or apprenticeship-related requirements where you intend to work.

Employer educational assistance

Employer tuition or education benefits can reduce out-of-pocket cost, but policies vary. Obtain written terms covering eligible programs, grades, reimbursement timing, required continued employment, repayment clauses, license or exam fees, tools, and continuing education before relying on reimbursement.

8. Apprenticeship and work-based learning

Apprenticeship is one of the central electrician pathways, but the term must be used precisely.

BLS identifies apprenticeship as the typical on-the-job training route for electricians. Apprenticeship programs may combine paid supervised work, progressive responsibility, wage progression, and related technical instruction.

At the same time, the current Apprenticeship.gov Occupation Finder page for O*NET **47-2111.00 Electricians** states that this occupation is **not currently approved for use in a Registered Apprenticeship Program** and provides an occupation-determination request path.

These two official-source facts require careful wording. The guide may accurately say apprenticeship is a common and typical pathway. It must not claim that the exact O*NET occupation is currently federally approved in the Occupation Finder without re-verification.

Before accepting an “apprenticeship,” verify:

- sponsor name and employer relationship;
- exact registered occupation and occupation code;
- registration authority and current status;
- whether the participant is an employee;
- starting wage and written wage progression;
- supervised work-hour requirements;
- related technical instruction;
- licensing or trainee-registration requirements;
- who pays for tuition, books, tools, PPE, testing, and travel;
- completion credential;

- cancellation, probation, and transfer rules.

A helper job, internship, pre-apprenticeship, employer training program, or private-school practicum may be valuable but should not be mislabeled as Registered Apprenticeship unless its status is verified.

9. Canada pathway

Occupation and wage evidence

Government of Canada Job Bank maps electricians (except industrial and power system) to **NOC 72200**. National wage data updated November 19, 2025 reported:

- **C\$20.00/hour low;**
- **C\$35.00/hour median;**
- **C\$48.00/hour high.**

These are national reference values using Job Bank/Statistics Canada data and are not guaranteed offers.

Red Seal — Construction Electrician

The Red Seal program identifies **Construction Electrician** as NOC 72200 and a Red Seal trade in participating provinces and territories. The occupational standard covers installing, altering, repairing, inspecting, verifying, commissioning, connecting, operating, maintaining, and decommissioning electrical systems across many environments.

Provincial and territorial apprenticeship authorities administer apprenticeship and certification. Harmonization materials recommend four technical-training levels, but implementation varies by jurisdiction.

Important: Red Seal endorsement demonstrates an interprovincial standard and supports mobility, but it does not erase province/territory-specific registration, compulsory-trade, permitting, contractor, inspection, or work-authorization rules. Verify the jurisdiction where you will train and work.

10. Colombia pathway

SENA electrical training

SENA provides public vocational pathways relevant to electrical work. Current source evidence includes:

- **Instalación de sistemas eléctricos residenciales y comerciales** offerings in Betowa; and
- a SENA program design for **Técnico Laboral en Electricidad Residencial, Comercial y de Sistemas Fotovoltaicos** with **2,400 hours** in the referenced 2025 resolution, divided into **1,536 lective hours** and **864 productive-stage hours**.

Programs, cohorts, locations, schedules, modality, and enrollment windows change. Recheck the current Betowa listing before relying on a particular city or date.

CONTE professional registration

The Consejo Nacional de Técnicos Electricistas (CONTE) states that it performs public functions that include studying, processing, and issuing **Matrículas Profesionales** for Técnicos Electricistas and maintaining the list of persons enabled for professional practice.

Training and professional authorization are not the same thing. Completing SENA training, private training, or employer experience must not be represented as automatically granting a CONTE professional registration or unrestricted legal authority.

Before paid professional practice in Colombia, verify directly with CONTE:

- the applicable registration category;
- education and experience evidence required;
- legal scope associated with the registration;
- application documents and current fees;
- whether expansion, special license, renewal, or other procedures apply;
- any additional RETIE, RETILAP, utility, employer, contractor, occupational-safety, or project requirements relevant to the work.

11. Latin America pathway

Electrical licensing, technical registration, vocational education, electrical codes, inspection systems, utility rules, and occupational-safety systems differ across Latin America.

For any country:

1. identify the national or subnational authority responsible for electrical-worker authorization;
2. check the official vocational-training and public-employment systems;
3. identify the adopted electrical code or technical regulation and enforcement authority;
4. verify apprenticeship, trainee, contractor, and professional-registration categories;
5. confirm whether foreign education or experience is recognized;
6. compare public/free training before paying a private provider;
7. verify the current job market through employer postings and public labor-market sources.

Do not transplant U.S. OSHA, Canadian Red Seal, Colombian CONTE, or another country's code/licensing rules into a different jurisdiction.

12. Skills to build deliberately

Technical foundations

Develop competence in:

- electrical theory: voltage, current, resistance, power, energy, AC/DC fundamentals, and basic circuit relationships;
- safe use of meters and test equipment under qualified supervision;
- conductor, raceway, device, panel, motor, transformer, grounding/bonding, and protective-device concepts appropriate to your level;
- electrical drawings, schematics, one-lines, panel schedules, symbols, specifications, and equipment labels;
- basic controls, relays, contactors, sensors, motor starters, and building-control concepts;
- troubleshooting using measured evidence instead of guesswork;
- preventive maintenance and inspection;
- code-navigation skills without pretending that memorization equals legal authority;
- work orders, digital documentation, photographs, test records, and CMMS platforms;
- renewable-energy, EV-charging, energy-storage, controls, or industrial concepts when relevant to the target market.

Professional skills

Develop:

- clear communication about hazards and uncertainty;
- accurate documentation;
- planning and material organization;
- teamwork across trades;
- customer service without making unsupported promises;
- punctuality and jobsite reliability;
- respect for permits, inspections, and supervision;
- willingness to report unsafe or noncompliant conditions instead of concealing them.

13. Tools, digital systems, cybersecurity, and privacy

Electrical work increasingly touches digital and networked systems. Depending on the role, technicians may encounter smart panels, building automation, variable-frequency drives, PLC-connected equipment, energy-management systems, solar monitoring, EV chargers, networked meters, access control, and remote diagnostics.

Treat digital access as part of the safety and security boundary.

- Use only employer-approved devices, applications, cables, adapters, and accounts.
- Do not connect personal laptops, USB devices, wireless hotspots, or diagnostic tools to customer or industrial networks without authorization.
- Protect credentials and multifactor-authentication methods.
- Do not photograph or upload panel schedules, network diagrams, security systems, facility layouts, credentials, proprietary drawings, or customer information to public services.
- Verify unusual remote commands, firmware changes, password resets, or configuration requests through approved channels.

- Escalate suspected compromise, unauthorized access, malware, abnormal control behavior, or unexpected network changes to the responsible IT/OT, controls, engineering, or security team.

Cybersecurity competence can be a career advantage, especially in industrial controls, data centers, critical facilities, smart buildings, renewable energy, and advanced maintenance, but it does not replace electrician qualifications.

14. Responsible use of artificial intelligence

AI can support low-risk learning and administrative tasks, but it is not a qualified electrician, inspector, engineer, licensing authority, code official, or safety professional.

Lower-risk uses

Using public or fictitious information, AI may help you:

- explain basic theory in different ways;
- generate study questions;
- practice apprenticeship or job interviews;
- create a learning schedule;
- improve grammar in a work-note example;
- practice reading fictional work orders;
- compare terminology across countries;
- organize a list of questions for a school, employer, licensing board, SENA, CONTE, or apprenticeship sponsor.

High-risk uses requiring authoritative verification

Do not rely on AI as sole authority for:

- energized-work procedures;
- absence-of-voltage verification;
- lockout/tagout decisions;
- conductor or raceway sizing;
- overcurrent or protective-device selection;
- grounding and bonding decisions;
- arc-flash or shock boundaries;
- hazardous-location classification;
- code compliance;
- permit or inspection decisions;
- electrical calculations used for real installation or safety decisions;
- legal licensing or professional-scope determinations;
- fault isolation where a wrong instruction could energize equipment or expose a person to hazardous energy.

AI can hallucinate code sections, formulas, equipment ratings, exceptions, and procedures. Verify real-world electrical decisions against current adopted codes, manufacturer information, employer procedures, qualified supervision, and the authority having jurisdiction.

15. Accessibility and accommodations

Electrical training and work can include physical, sensory, cognitive, communication, and environmental demands. People with disabilities may be able to perform many roles successfully with appropriate accommodations, task design, assistive technology, or specialization.

Helpful strategies may include:

- written procedures and checklists;
- diagrams paired with plain-language descriptions;
- speech-to-text or text-to-speech for documentation where safe and permitted;
- magnification and accessible digital training materials;
- structured labeling and tool organization;
- additional time or quiet conditions for written learning and examinations where legally available;
- visual, audible, or vibration alerts designed for the user's needs and the work environment;
- ergonomic tools or approved lifting aids;
- clear task sequencing and supervised practice.

Accommodation does not remove essential safety requirements or authorize a person to perform work outside training or legal scope. Discuss specific needs early with the training provider, employer, apprenticeship program, examination authority, or relevant accommodation office.

16. Job search, portfolio, and interview preparation

Resume evidence

Emphasize truthful evidence such as:

- supervised electrical training or apprenticeship status;
- relevant licenses, registrations, trainee cards, or credentials actually held;
- electrical safety and lockout/tagout training;
- installation or maintenance experience within verified scope;
- drawings, schematics, meters, tools, controls, motors, or systems actually used;
- work-order and CMMS documentation;
- construction, maintenance, industrial, facilities, solar, controls, HVAC, or adjacent trade experience;
- safety record and escalation judgment;
- customer service and teamwork.

Never claim “licensed electrician,” journeyman, master electrician, Red Seal, CONTE registration, high-voltage competence, apprenticeship completion, or specific code expertise unless you can support the claim.

Safe portfolio ideas

If employer and confidentiality rules allow, use:

- certificates or transcripts you legitimately earned;
- deenergized training-board projects;
- instructor-approved mock wiring diagrams;
- fictional troubleshooting exercises;
- photographs of training projects only when permitted and free of customer, location, security, or proprietary information;
- a preventive-maintenance checklist created from public manufacturer documentation;
- a short reflection explaining how you would stop and escalate an unsafe condition.

Never create a portfolio by performing unlicensed work, opening energized equipment for photographs, bypassing PPE, exposing customer information, or publishing proprietary building/security details.

Interview scenarios to practice

Prepare for questions such as:

- What do you do before working on equipment that may be energized?
- Tell me about a time you stopped work because conditions were unclear or unsafe.
- How do you verify that your meter or test method is appropriate?
- How do you handle a conflict between schedule pressure and a safety procedure?
- How do you document a fault you cannot fully resolve?
- What is your current license, trainee, apprentice, or registration status?
- Which electrical tasks are outside your present scope?
- How do you protect customer or facility information when using digital tools?

17. Twelve-week entry or advancement plan

Weeks 1-2 — Define the exact target

- Read the BLS Electricians profile and several real job postings.
- Decide whether your target is construction electrician, maintenance electrician, industrial electrician, residential electrician, electrical helper, apprentice, controls-focused role, or another adjacent path.
- Identify the licensing or trainee-registration authority where you intend to work.
- In Canada, identify the provincial/territorial apprenticeship authority.
- In Colombia, review current SENA offerings and CONTE requirements.

Weeks 3-4 — Build safe fundamentals

- Review electrical theory, basic mathematics, diagrams, and safety terminology.

- Practice only in deenergized or instructor-controlled environments.
- Learn lockout/tagout concepts and the difference between qualified and unqualified work.
- Build accurate tool and material vocabulary.

Weeks 5-6 — Compare training pathways

- Compare paid apprenticeship/trainee routes, public technical schools, community colleges, SENA, union/industry programs, and employer training.
- Verify licensing credit, recognized hours, tuition, tools, PPE, transportation, exam fees, and refund rules before paying.
- Ask whether prior experience can be formally evaluated.

Weeks 7-8 — Build job-search evidence

- Prepare a truthful resume.
- Gather legitimate training records and references.
- Practice safety, troubleshooting, and scope-boundary interview questions.
- Build a job/application tracker.

Weeks 9-10 — Verify funding and employer terms

Check:

- wages during training;
- wage progression;
- tuition and books;
- tools and PPE;
- licensing or registration costs;
- FAFSA or workforce eligibility when relevant;
- scholarships;
- employer educational assistance;
- repayment clauses;
- transportation and travel;
- overtime and shift expectations;
- accommodation processes.

Weeks 11-12 — Apply and refine

- Apply to verified roles that fit your current authorization and skills.
- Track which requirements repeatedly block progress.
- Improve one verified gap at a time.
- Avoid buying unrelated certificates simply to appear more qualified.
- Recheck licensing, apprenticeship, training, and job-market information before major spending decisions.

18. Verification checklist, sources, authorship, and license

Verify before enrolling or accepting work

- ☐ exact job duties and voltage/work environment;
- ☐ required apprentice, trainee, journeyman, master, contractor, professional, or other status;
- ☐ supervising person's required qualification;
- ☐ locally adopted electrical code and authority having jurisdiction;
- ☐ permit and inspection requirements;
- ☐ deenergization, LOTO, PPE, qualified-person, fall, confined-space, and other safety procedures;
- ☐ school/program recognition and whether training counts toward licensing or apprenticeship requirements;
- ☐ total tuition, tools, PPE, books, exam, licensing, travel, and transportation costs;
- ☐ funding or reimbursement eligibility in writing;
- ☐ wages, overtime, benefits, travel, per diem, and wage progression;
- ☐ current Canadian province/territory rules if applicable;
- ☐ current SENA availability and CONTE registration requirements in Colombia if applicable;
- ☐ employer cybersecurity and AI-use policy;
- ☐ accommodation process if needed.

Controlled sources reviewed for this edition

United States occupation and labor market

- U.S. Bureau of Labor Statistics — Electricians: <https://www.bls.gov/ooh/construction-and-extraction/electricians.htm>
- Indeed — Electrician salary (non-government market estimate): <https://www.indeed.com/career/electrician/salaries>

Apprenticeship and safety

- Apprenticeship.gov — Electricians occupation finder: <https://www.apprenticeship.gov/apprenticeship-occupations/listings?occupationCode=47-2111.00>
- OSHA — Electrical standards: <https://www.osha.gov/electrical/standards>
- OSHA — 1910 Subpart S: <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910SubpartS>
- OSHA — 1910.333 Selection and use of work practices: <https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.333>
- OSHA — Electrical contractors standards: <https://www.osha.gov/electrical-contractors/standards>

Canada

- Government of Canada Job Bank — Electrician wages, NOC 72200: <https://www.jobbank.gc.ca/wagereport/occupation/20684>
- Red Seal — Construction Electrician: <https://red-seal.ca/eng/trades/const-elect.shtml>
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Colombia

- SENA Betowa — Instalación de sistemas eléctricos residenciales y comerciales: <https://betowa.sena.edu.co/oferta/instalacion-de-sistemas-electricos-residenciales-y-comerciales?level=2&modality=P&offertype=company&programId=171095>
- SENA Normograma — Resolución 102057 de 2025 / program design evidence: https://normograma.sena.edu.co/compilacion/docs/resolucion_sena_2057_2025.htm
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- CONTE — institutional functions: <https://www.conte.org.co/nosotros/>

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Authorship and responsible AI use

Author: Alberto “Al” Leiva

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